

2, Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`.

Analyze the sequence of system calls and identify the kernel services involved

OUTPUT:

```
kannamina_abhishek@LAPTOP-D9F9IT80:~$ echo "Hello OSSP" > sample.txt
kannamina_abhishek@LAPTOP-D9F9IT80:~$ cat sample.txt
Hello OSSP
kannamina_abhishek@LAPTOP-D9F9IT80:~$ nano p2_copy.c
kannamina_abhishek@LAPTOP-D9F9IT80:~$ gcc p2_copy.c -o p2_copy
kannamina_abhishek@LAPTOP-D9F9IT80:~$ ./p2_copy
File copied successfully.
kannamina_abhishek@LAPTOP-D9F9IT80:~$ cat copy.txt
```

CODE:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int src, dest;
    char buffer[1024];
    ssize_t n;

    src = open("sample.txt", O_RDONLY);
    if (src < 0) {
        perror("open source");
        return 1;
    }

    dest = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest < 0) {
        perror("open destination");
        close(src);
        return 1;
    }

    while ((n = read(src, buffer, sizeof(buffer))) > 0) {
        write(dest, buffer, n);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```